



Kinesthesia: A neuroplasticity-based cognitive–motor intervention for brain health in aging—proof-of-concept pilot evidence

ARTICLE INFO

Keywords:

Healthy aging
Cognitive–motor integration
Neuroplasticity
Exergames
Mild cognitive impairment
Dual-task training
Brain stimulation

ABSTRACT

Background: Cognitive and functional decline in older adults poses a growing global health challenge. In the absence of curative pharmacological treatments for dementia, non-pharmacological strategies combining physical and cognitive stimulation are increasingly prioritised. Simultaneous cognitive–motor training is theoretically grounded in the interdependence of motor and cognitive networks and has shown promise in meta-analytic evidence, yet scalable, neuroplasticity-coherent platforms designed from first principles remain scarce.

Objectives: To describe the conceptual architecture, neurophysiological rationale, and neuroplasticity-based design principles of Kinesthesia, and to report preliminary feasibility and proof-of-concept data from an uncontrolled pilot study in older adults with and without Mild Cognitive Impairment (MCI).

Methods: Twenty-two older adults (11 cognitively healthy, 11 MCI) completed 30 sessions ($3 \times$ /week, 10 weeks) of Kinesthesia, a therapeutic videogame platform integrating full-body kinematic interaction with progressive cognitive–motor demands structured around six established neuroplasticity principles. Outcomes included neuropsychological assessment (NEUROPSI Atención y Memoria), functional walking capacity (6-Minute Walk Test), balance and gait (Tinetti, TUG), within-platform learning trajectories, and Spearman correlations between game learning indices and clinical change scores.

Results: Adherence was 100%, and satisfaction was uniformly high. Both subgroups showed significant pre-to-post improvements in delayed memory, executive functions, and functional walking capacity, with predominantly large effect sizes ($d = 0.5$ – 1.4). Within-platform learning was systematic across games ($\geq 80\%$ of participants improving per game). Spearman correlations revealed that learning in foundational locomotor games (Double Steps, Follow Tiles) predicted clinical cognitive gains most robustly in the combined sample, supporting a cognitive–motor scaffolding hypothesis.

Conclusions: Kinesthesia demonstrated feasibility and preliminary proof-of-concept as a scalable, affordable, neuroplasticity-grounded cognitive–motor intervention for healthy aging. All findings are hypothesis-generating and require confirmation in adequately powered randomised controlled trials with active comparators.

Dear Editor,

Cognitive decline in aging is a leading cause of disability and dependence, yet scalable, evidence-based non-pharmacological interventions combining cognitive and motor stimulation remain scarce, particularly in low- and middle-income countries (LMIC) (GBD, 2022; Livingston et al., 2020). We present Kinesthesia, a therapeutic videogame platform in which cognitive and motor demands are structurally inseparable within each task. Unlike conventional dual-task protocols or post-hoc exergame adaptations, Kinesthesia was designed from first principles to operationalize simultaneously all six neuroplasticity prerequisites identified by Nahum et al. (2013). In brief: (1) Attention—simultaneous cognitive–motor demands requiring continuous undivided attention, with immediate failure as intrinsic motivation; (2) Accuracy—centimeter-resolution motion capture enforcing spatially and temporally precise responses; (3) Adequacy—a bidirectional adaptive difficulty algorithm operating in real time within and across sessions; (4) Reinforcement—immediate visual and auditory feedback on performance outcome embedded in every game; (5) Frequency—30 sessions of 60 min, $3 \times$ /week over 10 weeks, intensity

reinforces neuroplasticity; (6) Flexibility—modular game suite (13 games across 5 cognitive–motor categories) with individualized session composition through structured whole-body interaction including contralateral midline-crossing movements (Chiarenza & Bosch-Bayard, 2025).

Of 70 older adults screened, 22 met all inclusion criteria and completed the full protocol (exclusion grounds: medical comorbidities, functional mobility threshold, cognitive criteria outside the Healthy Control (HC) – Mild Cognitive Deterioration (MCI) spectrum, and neuropsychiatric contraindications). Only people who were classified as Healthy (HC) or MCI in both MMSE and MoCA were included. We conducted a single-arm pre–post pilot study at the Centro Gerontológico de Día “Plan Vida”, Querétaro, Mexico (2017). Twenty-two community-dwelling older adults (mean age 69.7 ± 3.6 years; 86.4% women) completed 30 Kinesthesia sessions (60 min, $3 \times$ /week, 10 weeks). The cognitively healthy subgroup (HC, $n = 11$) had a mean age of 70.1 ± 3.8 years, 90.9% women, and education 13.9 ± 2.5 years (MoCA 27.4 ± 0.8); the MCI subgroup ($n = 11$) had a mean age of 69.3 ± 3.4 years, 81.8% women, and education 7.6 ± 3.0 years (MoCA 21.6 ± 3.7). Sex

<https://doi.org/10.1016/j.archger.2026.106346>

Received 3 May 2026; Received in revised form 18 June 2026; Accepted 24 June 2026

Available online 25 June 2026

0167-4943/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

imbalance was comparable across subgroups (sample-level characteristic); subgroups differed substantially in education ($p = 0.0001$). MCI was defined as MoCA < 26 (cognitive reserve stratifier; formal neuropsychological confirmation was not performed—acknowledged as a limitation). Outcomes included the NEUROPSI Atención y Memoria neuropsychological battery (Ostrosky-Solis et al., 2007) (age- and education-stratified norms for Spanish-speaking populations; normalised score scale: ≤ 69 severe impairment, 70–84 mild impairment, 85–115 normal, ≥ 116 high normal; no ceiling effects), the Six-Minute Walk Test (6MWT; benchmarked against Enright age-stratified normative values (Enright & Sherrill, 1998). All procedures complied with the Declaration of Helsinki; ethics approval was obtained from the Faculty of Medicine, Universidad Autónoma de Querétaro (protocols 02-03/11/2016 and 03-01/12/2016, Grant FOPER No 143834). Pre-to-post comparisons used sign-flip permutation tests (5000 permutations) with Benjamini–Hochberg false discovery rate correction within each instrument domain.

Adherence was 100% and satisfaction ratings were uniformly high (mean 4.75/5.0) in both subgroups. Both groups showed significant pre-to-post improvements in delayed memory and executive functions (Table 1). In the cognitively healthy group, global normalised NEUROPSI score improved from a mean of 108.3 to 116.5 ($d = 1.38$, $p = 0.002$), crossing the normative threshold from “normal” into “high normal” range. In the MCI group, delayed verbal recall and associated pairs evocation showed large effects ($d \geq 0.75$) that reached significance in the combined-sample analysis ($p \leq 0.010$, $r \geq 0.59$)—domains particularly sensitive to early cognitive decline and for which gains in unimodal interventions are typically modest (Gallou-Guyot et al., 2020; Karssemeijer et al., 2017).

Functional walking capacity improved substantially in both subgroups: mean 6MWT distance increased from 433.1 ± 77.7 m to 495.8 ± 49.0 m ($+62.6$ m; $d = 1.06$, $p = 0.006$) in the cognitively healthy group and $+57.6$ m ($d = 1.40$, $p = 0.004$) in the MCI group (baseline 396.2 ± 71.3 m $\rightarrow 453.8 \pm 63.0$ m). Critically, the theoretical percentage of the age-expected normative value (Enright equation) improved from 78.4% to 90.9% in the healthy group and from 76.7% to 84.3% in the MCI group—approaching but not reaching normative levels from a substantially subnormal baseline (combined pre: 415.6 ± 75.1 m; post: 475.9 ± 58.5 m). Because both NEUROPSI and the 6MWT provide age-stratified normative reference values, the observed improvements can be benchmarked against the expected performance of a healthy same-age population, providing an objective measure of clinical magnitude. This normative contextualization complements but does not substitute for a

controlled experimental design.

Spearman correlations between within-platform learning indices and clinical change scores revealed that mastery of the two most foundational bilateral locomotor games—Two Steps (bilateral lower-limb dissociation to independent spatial targets with real-time foot-target assignment) and Follow the Path (full-body reproduction of a memorised spatial route)—was most consistently associated with cognitive gains across the combined sample ($\rho = 0.50$ – 0.80 , $p < 0.01$), supporting a cognitive–motor scaffolding hypothesis—the possibility that whole-body sensorimotor integration at a foundational level may support higher-order cognitive gains—but these associations are exploratory and do not establish directionality or mechanism. A full effect-size summary is presented in Table 1.

These findings provide initial proof of concept for Kinesthesia as a feasible, acceptable, and potentially efficacious cognitive–motor intervention for older adults across the cognitive spectrum. The pattern of large effects in delayed memory and robust functional walking gains is consistent with the theoretical engagement of frontoparietal, cortico-cerebellar, and basal ganglia networks through embodied cognitive–motor interaction (Leisman et al., 2016)—a mechanistic account that remains to be confirmed by direct neurophysiological measurement. Resting-state EEG was additionally recorded in the same cohort, providing an independent neurophysiological dataset to test these mechanistic accounts directly that will be published separately. Kinesthesia operates on standard hardware with affordable motion-capture sensors, is designed for clinical, community, and home-based deployment, and is particularly well-suited to LMIC settings where scalable, low-cost brain health tools are most urgently needed (Livingston et al., 2020; Wollesen & Voelcker-Rehage, 2014). These results provide the power estimates and rationale for a confirmatory randomized controlled trial with an active comparator.

The primary limitation is the absence of a control group, which precludes causal attribution; practice effects and regression to the mean cannot be excluded. Cognitive subgroup classification used dual MMSE/MoCA screening without formal neuropsychological confirmation or neuroimaging—noted as a limitation, though convergent dual-instrument classification provides a clinically validated basis for stratification. Additional limitations include the small sample ($N = 22$), single-centre design, non-probabilistic sampling, gender imbalance (86.4% women), and absence of follow-up data. Data collection was completed in 2017 as part of two master's thesis projects (Grant FOPER No 143834, UAQ) within a strictly controlled protocol; formal publication was delayed by resource and institutional constraints, during

Table 1

Primary outcomes: pre-to-post changes by subgroup and combined sample. Pre M $\pm$ SD and post M $\pm$ SD = group means $\pm$ SD; Δ = mean change; d = Cohen's d ; p = permutation sign-flip test (5000 permutations); q (BH) = Benjamini–Hochberg FDR-corrected p -value within each instrument domain. Green rows = BH-significant combined-sample result.

Outcome measure	Group	Pre M $\pm$ SD	Post M $\pm$ SD	Δ	d	p	q (BH)
Motor outcome							
6MWT metres walked	HC ($n = 10$)	433.1 $\pm$ 77.7	495.8 $\pm$ 49.0	+62.6	1.06	0.006	0.024
	MCI ($n = 9$)	396.2 $\pm$ 71.3	453.8 $\pm$ 63.0	+57.6	1.40	0.004	0.024
	Combined ($N = 19$)	415.6 $\pm$ 75.1	475.9 $\pm$ 58.5	+60.2	1.20	<0.001	0.002
6MWT % theoretical	HC ($n = 10$)	78.4 $\pm$ 15.0	90.9 $\pm$ 8.6	+12.5	1.11	0.006	0.024
	MCI ($n = 9$)	76.7 $\pm$ 13.5	84.3 $\pm$ 15.4	+7.6	0.48	0.129	—
	Combined ($N = 19$)	77.6 $\pm$ 13.9	87.8 $\pm$ 12.4	+10.2	0.76	0.002	0.023
Cognitive outcomes (NEUROPSI Atención y Memoria)							
Global normalized score	HC ($n = 10$)	108.3 $\pm$ 13.4	116.5 $\pm$ 10.8	+8.2	1.38	0.002	0.012
	MCI ($n = 9$)	102.2 $\pm$ 14.5	102.8 $\pm$ 11.5	+0.6	0.05	0.867	—
	Combined ($N = 20$)	105.4 $\pm$ 13.9	110.0 $\pm$ 12.9	+4.6	0.93	<0.001	0.008
Verbal recall (spontaneous)	Combined ($N = 20$)	6.63 $\pm$ 2.75	8.26 $\pm$ 2.22	+1.63	1.07	<0.001	0.004
Assoc. pairs evocation	Combined ($N = 20$)	8.16 $\pm$ 2.9	9.74 $\pm$ 2.4	+1.58	0.81	0.010	0.023
Rey figure evocation	Combined ($N = 20$)	11.8 $\pm$ 6.2	13.8 $\pm$ 7.3	+2.00	0.64	0.013	0.052
Assoc. pairs evocation	Combined ($N = 20$)	8.16 $\pm$ 2.9	9.74 $\pm$ 2.4	+1.58	0.81	0.010	0.023

Colour scale: ■ large effect ($d \geq 0.80$ / $r \geq 0.50$); ■ medium-large ($0.50 \leq d < 0.80$); ■ medium ($0.30 \leq d < 0.50$); white = small/ns. Significance: ** $p < 0.01$; * $p < 0.05$; †large effect, $p < 0.10$ (trend). ns = not significant; = not analyzed or ceiling effect. 6MWT = Six-Minute Walk Test; MCI = Mild Cognitive Impairment; NEUROPSI = NEUROPSI Atención y Memoria battery. NEUROPSI and 6MWT are benchmarked against published age-stratified norms (see text); improvements in these measures are interpretable against an objective normative reference that does not substitute for a controlled experimental design.

which the platform has been deployed in additional settings in Mexico and Italy (since 2024) and extended to children with neurodevelopmental disorders. The present report represents the formal publication of the original pilot dataset. Resting-state EEG was additionally recorded in the same cohort, providing the basis for a separate neurophysiological analysis of state-dependent electrocortical reorganisation and its relationship to baseline cognitive status and individual game-learning trajectories. A pre-registered RCT with an active comparator is the necessary next step, incorporating neurophysiological assessments in a larger cohort to establish whether these state-dependent EEG signatures replicate and whether electrocortical change mediates the behavioral gains reported here.

CRedit authorship contribution statement

Jorge Bosch-Bayard: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Judith Guerrero-Sauzamedia:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Laura Navarrete-Torres:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Rodolfo Bosch-Bayard:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Giuseppe A. Chiarenza:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Liane Aguilar-Fabre:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **René Rodríguez-Valdés:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Chiarenza, G. A., & Bosch-Bayard, J. (2025). Utilizing movement-related potentials to monitor cognitive rehabilitation in individuals with central nervous system

- disorders: A review. *International Journal of Psychophysiology: Official Journal of the International Organization of Psychophysiology*, 208, Article 112485. <https://doi.org/10.1016/j.ijpsycho.2024.112485>
- Enright, P. L., & Sherrill, D. L. (1998). Reference equations for the six-minute walk in healthy adults. *American Journal of Respiratory and Critical Care Medicine*, 158(5 Pt 1), 1384–1387. <https://doi.org/10.1164/ajrcm.158.5.9710086>
- Gallou-Guyot, M., Mandigout, S., Bherer, L., & Perrochon, A. (2020). Effects of exergames and cognitive-motor dual-task training on cognitive, physical and dual-task functions in cognitively healthy older adults: an overview. *Ageing Research Reviews*, 63, 101135. <https://doi.org/10.1016/j.arr.2020.101135>
- GBD 2019 Dementia Forecasting Collaborators. (2022). Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050: An analysis for the Global Burden of Disease Study 2019. *Lancet Public Health* (pp. e105–e125). [https://doi.org/10.1016/S2468-2667\(21\)00249-8](https://doi.org/10.1016/S2468-2667(21)00249-8), 7(2).
- Karssemeijer, E. G. A., Aaronson, J. A., Bossers, W. J. R., et al. (2017). Positive effects of combined cognitive and physical exercise training on cognitive function in older adults with mild cognitive impairment or dementia: a meta-analysis. *Ageing Research Reviews*, 40, 75–83. <https://doi.org/10.1016/j.arr.2017.09.003>
- Leisman, G., Moustafa, A. A., & Shafir, T. (2016). Thinking, walking, talking: Integratory motor and cognitive brain function. *Frontiers in Public Health*, 4, 94. <https://doi.org/10.3389/fpubh.2016.00094>
- Livingston, G., Huntley, J., Sommerlad, A., Ames, D., Ballard, C., Banerjee, S., et al. (2020). Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*, 396(10248), 413–446. [https://doi.org/10.1016/S0140-6736\(20\)30367-6](https://doi.org/10.1016/S0140-6736(20)30367-6)
- Nahum, M., Lee, H., & Merzenich, M. M. (2013). Principles of neuroplasticity-based rehabilitation. *Progress in Brain Research*, 207, 141–171. <https://doi.org/10.1016/B978-0-444-63327-9.00009-6>
- Ostrosky-Solís, F., Gómez-Pérez, M. E., Matute, E., Rosselli, M., Ardila, A., & Pineda, D. (2007). NEUROPSI ATTENTION AND MEMORY: A neuropsychological test battery in Spanish with norms by age and educational level. *Applied Neuropsychology*, 14(3), 156–170. <https://doi.org/10.1080/09084280701508655>
- Wollesen, B., & Voelcker-Rehage, C. (2014). Training effects on motor–cognitive dual-task performance in older adults: A systematic review. *European Review of Aging and Physical Activity*, 11(1), 5–24. <https://doi.org/10.1007/s11556-013-0122-z>

Jorge Bosch-Bayard^{a,b,c,1,*}, Judith Guerrero-Sauzamedia^{a,b,1},
Laura Navarrete-Torres^{a,1}, Rodolfo Bosch-Bayard^{d,e}, Giuseppe
A. Chiarenza^f, Liane Aguilar-Fabre^a, René Rodríguez-Valdés^g

^a Clínica del Sistema Nervioso, Universidad Autónoma de Querétaro,
México

^b Centro Integrador de la Mente, la Conducta y el Movimiento (CIMMCO),
Querétaro, México

^c Carl von Ossietzky Universität Oldenburg, Germany

^d State University of Zanzibar, Tanzania

^e Universidad de Ciencias Médicas de La Habana, Cuba

^f CIDAAl, Milano, Italy

^g Vithas Málaga, Grupo Hospitalario Vithas, Málaga 29016, Spain

* Corresponding author at: Department of Psychology, Universität
Oldenburg, Oldenburg, 26129, Germany.

E-mail address: oldgandalf@gmail.com (J. Bosch-Bayard).

¹ Equal contribution as first authors.